

THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE

DEPARTMENT OF CHEMISTRY

SECOND SEMESTER CBE

2018/19 SESSION

COURSE CODE: CHE 102

COURSE TITLE: GENERAL CHEMISTRY II

INSTRUCTIONS:

A. ANSWER SIXTY QUESTIONS (60) ONLY

B. THERE ARE FIVE (5) SECTIONS, ANSWER AT LEAST TWELVE (12) QUESTIONS FROM EACH SECTION

TIME ALLOWED: 30 MINS

SECTION 1

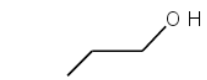
1. Which of the following is otherwise known as grain alcohol?
 - a) Propanol
 - b) Butanol
 - c) Ethanol+++
 - d) Methanol
2. Alcohols have higher boiling points than do alkanes of similar molecular weight because
 - a) Oxygen is present in alcohols which is deficient in alkanes
 - b) Strong ionic bonding is present in alcohols while covalent bonding is present in alkanes
 - c) Interaction between oxygen of the hydroxyl group with other molecules of alcohols which is absent in alkanes+++
 - d) Alkanes are largely hydrocarbons which mostly are in gaseous states
3. The carbon atom bonded to a hydroxyl group is called

- a) carbol
 - b) carbocation
 - c) carbinol+++
 - d) carbene
4. Glycols have———— hydroxyl group(s).
- a) one
 - b) two+++
 - c) three
 - d) four
5. Which of the alcohols is otherwise known as wood spirit?
- a) Ethanol
 - b) Propanol
 - c) Methanol+++
 - d) Butanol
6. Which of these is not another name for 2-propanol?
- a) Isopropanol
 - b) Isopropyl alcohol
 - c) Rubbing alcohol
 - d) Vinyl propanol+++
7. Alcohols react with active metals (Na, K, etc.) to produce ————gas.
- a) oxygen
 - b) hydrogen+++
 - c) nitrogen
 - d) helium
8. Which of these chemicals would you recommend for a person who has a dry skin?
- a) Ethane 1, 2-diol
 - b) Propane 1, 2, 3-triol+++
 - c) Glycol
 - d) Glyoxal
9. The green engine coolant in an automobile radiator will likely contain one of these

- a) Glycerol
- b) Ethane 1, 2-diol+++
- c) Ethylene
- d) 2-propanol

10. An example of a polyhydric alcohol is

- a) Glycerol
- b) Glycol
- c) Cellulose+++
- d) Polymer

11. The formation of ethylene glycol () from ethylene (=) is an example of

- a) Oxidation+++
- b) reduction
- c) dehydration
- d) hydrogenation

12. Which of these is a secondary alcohol?

- a) $\text{CH}_3\text{CH}_2\text{OH}$
- b) $\text{CH}_3(\text{CH}_3)\text{CHOH}$ +++
- c) $\text{CH}_3(\text{CH}_3)_2\text{COH}$
- d) $\text{CH}_3\text{CH}_2\text{COH}$

13. An example of a compound that contains a carbonyl group is

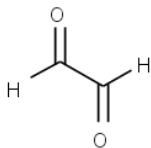
- a) H_2CO +++
- b) CH_3OH
- c) CH_2CH_2
- d) CH_3OCH_3

14. One mole of NaBH_4 reacts with 1 mole of acetone, $\text{CH}_3\text{C}(\text{O})\text{CH}_3$, to give, after treatment with aqueous acid,—

- a) Propanoic acid
- b) 2-propanone
- c) 2-propanol+++
- d) 2-propanal

15. Which of these is not true about alcohols?

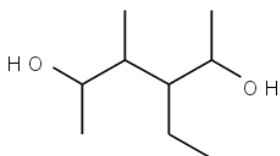
- a) The melting and boiling points of butanol is higher than ethanol
- b) Ethanol is more soluble in water than octanol
- c) Hexanol is more miscible with water than Methanol+++
- d) The boiling point of methanol is higher than that of ethane



16. LAH reacts with glyoxal, , to give

- a) $\text{CH}_3\text{CO}_2\text{H}$
- b) $\text{HOCH}_2\text{CH}_2\text{OH}+++$
- c) H_2CCH_2
- d) $\text{HOCH}_2\text{CO}_2\text{H}$

17. The correct name for the structure drawn is



- a) 4-isopropoxyl-3-methyl hexan-2-ol
- b) 4-ethyl-3-methyl-hexan-2, 5-diol+++
- c) Nonan-2, 5- diol
- d) 4-ethyl-2, 3, 5-trimethyl hexan-1, 6-diol

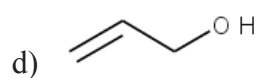
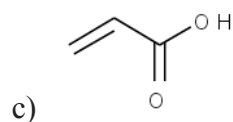
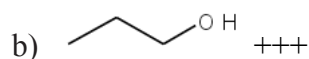
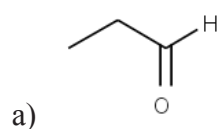
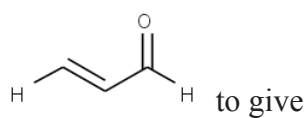
18. Which of these will not mix with water in all proportions?

- a) Ethanol
- b) Methanol
- c) β -butanol
- d) Nonanol+++

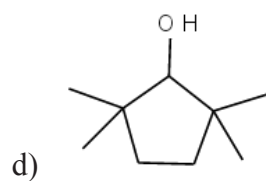
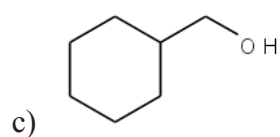
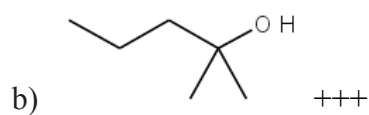
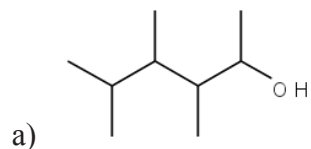
19. LAH reacts with $\text{H}_2\text{CCHC}(\text{O})\text{H}$ to give

- a) $\text{CH}_3\text{CH}_2\text{C}(\text{O})\text{H}$
- b) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
- c) $\text{H}_2\text{CCHCO}_2\text{H}$
- d) $\text{H}_2\text{CCHCH}_2\text{OH}+++$

20. Hydrogen (H_2), in the presence of Raney Ni catalyst, reacts with



21. Which of these is a tertiary alcohol?



22. Alcohol can be prepared by all the following ways except

- a) Addition of acid to alkene
- b) Addition of bromine water to alkene
- c) Addition of borane tetrahydrofuran to alkene followed by alkaline hydrogen peroxide
- d) Addition of Grignard reagent to alkene+++

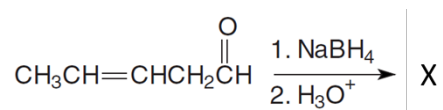
23. Which of these compounds has no carbonyl group present in it?

- a) Alkanals
- b) Amides

- c) Acid chlorides
 - d) Nitriles+++
24. In a carbonyl group, the oxygen atom is said to be
- a) Electrophilic and reacts with nucleophiles
 - b) Nucleophilic and reacts with electrophiles+++
 - c) Electronegative and reacts with bases
 - d) Nucleophilic and reacts with bases
25. In the reduction of ketone to a secondary alcohol by either NaBH_4 or LAH, why will you prefer not to add aqueous acid in the first step?
- a) It will reduce a ketone to a primary alcohol instead
 - b) There will be no reaction
 - c) Acid will have reacted with the reagent before the carbonyl has any chance to undergo reduction+++
 - d) It causes a serious explosion that can set the whole laboratory on fire
26. Reaction involving the use of lithium aluminium hydride (LAH) must always be in inert solvents because
- a) LAH is inactive in any other solvents
 - b) The presence of reactive solvents will produce large amount of side reaction products
 - c) LAH will react with solvents having acidic hydrogen atoms+++
 - d) LAH is very reactive and catches fire easily
27. Why will you prefer LAH to NaBH_4 as reducing agent for carbonyl group?
- a) LAH is cheaper than NaBH_4
 - b) NaBH_4 is too slow to reduce carbonyl group+++
 - c) NaBH_4 is not as selective as LAH
 - d) LAH is more environment-friendly than LAH
28. What are the products of the reaction of $\text{CH}_3\text{CH}_2\text{C}(\text{O})\text{OCH}_3$ with LAH followed by treatment with acid?
- a) 1-propanol and methanol+++
 - b) Propanal and methanol
 - c) Ethanol and ethanoic acid
 - d) 2-propanol and methanol
29. Which of these reagents will give a saturated alcohol with an unsaturated aldehyde?

- a) $\text{NaBH}_4/\text{H}_3\text{O}^+$
- b) $\text{LAH}/\text{H}_3\text{O}^+$
- c) $\text{H}_2/\text{Raney Nickel}+++$
- d) CH_3MgBr

30. What is X in this reaction?



- a) $\text{CH}_3\text{CHCHCH}_2\text{COH}+++$
- b) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{COH}$
- c) $\text{CH}_3\text{CHCH}_2\text{CH}_2\text{COH}$
- d) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHCOH}$

31. What is the function of H_3O^+ in the reaction of RMgX with carbonyl compounds?

- a) To produce magnesium salt of an alkoxide ion
- b) To convert the alkoxide ion to alcohol+++
- c) To neutralize the alkaline reaction medium
- d) To reduce the carbonyl group to alcohol

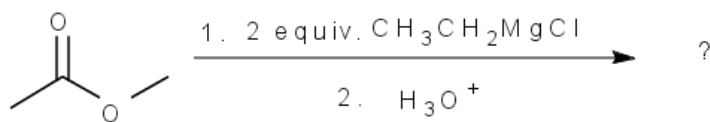
32. Reactions involving RMgX must be carried out in inert solvents like ether because

- a) Solvents having acidic protons will react with and destroy the reagent before it could react with the carbonyl compound+++
- b) Reactive solvents will interact with the reagent to form undesired products with the carbonyl compounds
- c) RMgX , being a form of salt, is very reactive toward solvents having acidic protons
- d) RMgX is a very sensitive reagent and becomes deactivated upon contact with reactive solvents.

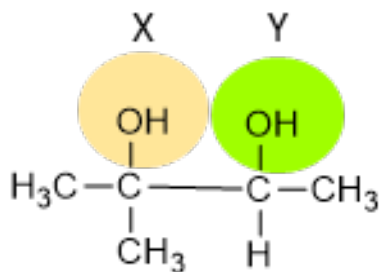
33. Which of these groups does not possess an acidic proton?

- a) $-\text{C}\equiv\text{CH}$
- b) $-\text{OH}$
- c) $-\text{O}-\text{CH}_3+++$
- d) $-\text{NH}_2$

34. What is(are) the product(s) of this reaction?



- a) 3-methyl-3-pentanone
 - b) 3-methyl-3-pentanone and methanol
 - c) 3-methylpentan-3-ol and methanol+++
 - d) 3-methylpentan-3-ol
35. What is the product of the reaction of 1 equivalent of a Grignard reagent with an ester, acyl chloride or carboxylate salt?
- a) Aldehyde
 - b) Alkanone+++
 - c) Alkanoic acid
 - d) Alcohol
36. Which of these reagents will oxidize primary alcohols directly to aldehydes?
- a) $\text{Cr}_2\text{O}_7^{2-}$
 - b) MnO_4^-
 - c) PCC+++
 - d) HIO_4
37. Oxidation of secondary alcohols in either acidic or basic conditions will give
- a) Aldehydes
 - b) Ketones+++
 - c) Ozonides
 - d) Carboxylic acids
38. What is the product you would obtain if you oxidize tertiary alcohol in either dichromate or permanganate?
- a) Alkanone only
 - b) No product+++
 - c) Alkanone and Alkanal
 - d) Alkanal only
39. Identify X and Y as different classes of alcohols present in the structure drawn



- a) X is 2° alcohol, Y is 3° alcohol
 - b) X is 2° alcohol, Y is 1° alcohol
 - c) X is 3° alcohol, Y is 2° alcohol+++
 - d) X is 3° alcohol, Y is 1° alcohol
40. A terminal ketone is otherwise known as
- a) Aldehyde+++
 - b) Alkanone
 - c) Alkanoic acid
 - d) Freon
41. The carbonyl group is capable of undergoing the following reactions except
- a) Oxidation
 - b) Reduction
 - c) Nucleophilic addition
 - d) Elimination+++
42. The preservative used for biological specimens contains
- a) Ethanol
 - b) Methanal+++
 - c) Formic acid
 - d) Trioxane
43. To prepare formalin I would dissolve
- a) 73 g formaldehyde in 17 g water
 - b) 37 g formaldehyde in 63 g water+++
 - c) 37 g formaldehyde in 100 g water
 - d) 63 g formaldehyde in 37 g water
44. A pure source of methanal is
- a) Dry formaldehyde
 - b) Trioxane+++

- c) Dioxane
 - d) Acetone
45. Acetone is used in organic laboratories for cleaning glassware because it is
- a) Very cheap
 - b) Highly volatile and polar+++
 - c) Colourless and non-viscous
 - d) Colourless and soapy
46. Which of these could wood varnishes contain?
- a) Propanol
 - b) 2-butanone+++
 - c) Benzene
 - d) Ethanol
47. CH_3CHO reacts first with ethyl lithium ($\text{CH}_3\text{CH}_2\text{Li}$) and then with dilute acid to give
- a) ethane
 - b) butan-2-ol+++
 - c) butane
 - d) no reaction
48. Lithium reacts with *n*-butyl chloride to give
- a) *n*-butane
 - b) *n*-butyllithium+++
 - c) chlorine
 - d) hydrogen
49. Formaldehyde (HCHO) reacts first with CH_3MgCl and then with dilute acid to give a
- a) primary alcohol+++
 - b) secondary alcohol
 - c) tertiary alcohol
 - d) quaternary alcohol
50. The oxidation of $\text{CH}_3\text{CH}_2\text{OH}$ with acidic dichromate ($\text{Cr}_2\text{O}_7^{2-}$) gives
- a) a ketone
 - b) ethyl alcohol
 - c) an aldehyde

d) an acid+++

51. $\text{CH}_3\text{CH}_2\text{OH}$ reacts with pyridinium chlorochromate (PCC) to give

- a) a ketone
- b) an alcohol
- c) an aldehyde+++
- d) an acid

52. PBr_3 reacts with ethanol to give

- a) tribromoethane
- b) bromoethanol
- c) bromoethane+++
- d) no reaction

53. Thionyl chloride, ClS(O)Cl , reacts with isopropyl alcohol, $(\text{CH}_3)_2\text{CHOH}$, to give

- a) 2-chloropropane+++
- b) 1-chloropropane
- c) 2-propanol
- d) acetone (a ketone)

54. Isopropyl alcohol, $(\text{CH}_3)_2\text{CHOH}$, reacts with hot concentrated H_2SO_4 to give

- a) Propane
- b) 2-propanol
- c) Propene+++
- d) Propyne

55. Aldehydes have ——— boiling points than/as alcohols of comparable molecular weight.

- a) higher
- b) lower+++
- c) about the same
- d) exactly the same

56. Acetone is the common name for

- a) 1-butanone
- b) 1-propanone
- c) 2-propanone+++
- d) 2-butanone

57. Which material is most reactive toward nucleophilic addition?
- a) Acetaldehyde+++
 - b) dimethyl ketone
 - c) diisopropyl ketone
 - d) di-*tert*-butyl ketone
58. The oxidation of 1-propanol with chromic acid yields
- a) an alcohol
 - b) an aldehyde
 - c) a ketone
 - d) an acid+++
59. The oxidation of 2-propanol with chromic acid yields
- a) an alcohol
 - b) an aldehyde
 - c) a ketone+++
 - d) an acid
60. The oxidation of 1-propanol with PCC gives
- a) an alcohol
 - b) an aldehyde+++
 - c) a ketone
 - d) an acid
61. The oxidation of ——— with ozone followed by reduction with $(\text{CH}_3)_2\text{S}$ yields one product, 1,5-pentanedial.
- a) 1,5-pentanediol
 - b) 2,4-pentanedione
 - c) Cyclopentene+++
 - d) cyclopentane
62. The acid-catalyzed hydration of cyclodecyne yields
- a) Cyclodecanone+++
 - b) cyclodecadiene
 - c) cyclodecanediol
 - d) cyclodecanedial
63. Reduction of acetic acid with $\text{Li}(\text{tri-}i\text{tert-butoxy})\text{AlH}$ yields
- a) an acid
 - b) an aldehyde+++

- c) an alcohol
 - d) a ketone
64. The reaction of acetyl chloride with excess Grignard reagent followed by acidification yields
- a) a ketone
 - b) an ester
 - c) a secondary alcohol
 - d) a tertiary alcohol+++
65. Lithium dimethylcuprate (Gilman reagent) reacts with acetyl chloride to give
- a) a ketone+++
 - b) an ester
 - c) a secondary alcohol
 - d) a tertiary alcohol
66. The cyanide ion $\text{C}\equiv\text{N}^-$ reacts with a ketone to give
- a) an acid
 - b) a cyanohydrin+++
 - c) an aldehyde
 - d) an amine
67. The acid-catalyzed addition of water to formaldehyde gives
- a) a ketal
 - b) a hemiacetal
 - c) an acetal
 - d) a hydrate+++
68. The acid-catalyzed addition of 2 mol of alcohol to 1 mol of aldehyde yields
- a) a ketal
 - b) a hemiacetal
 - c) an acetal+++
 - d) a hydrate
69. Ammonia reacts with an aldehyde to give an
- a) alcohol
 - b) amide
 - c) imine+++
 - d) amine
70. The reaction of a Wittig reagent with ketones give

- a) Alkenes+++
- b) ketones
- c) aldehydes
- d) alkanes

71. Silver ion, Ag^+ , reacts with aldehydes to form

- a) a red precipitate
- b) a silver “mirror”+++
- c) a blue solution
- d) there is no reaction

72. Dimethylamine reacts with a ketone to give an

- a) amide
- b) imide
- c) amine
- d) enamine+++

73. Carbocations are

- a) carbon atoms with a negative charge
- b) carbon atoms with a positive charge+++
- c) carbon atoms with no charge
- d) any atom with a positive charge

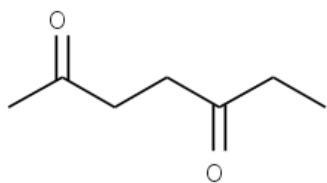
74. A—— carbocation has the greatest stability.

- a) methyl
- b) primary
- c) secondary
- d) tertiary+++

75. Carbocations are stabilized by adjacent groups that—— the carbocation.

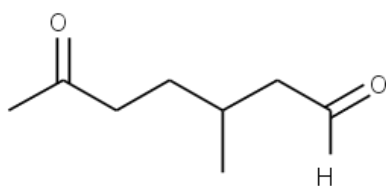
- a) donate electrons to+++
- b) withdraw electrons from
- c) donate protons to
- d) withdraw protons from

76. What is the name of this compound?



- a) 2,5-heptanedione+++
- b) 2,5-heptadione
- c) 3,6-heptadione
- d) 3,6-heptanedione

77. The correct name for this compound is



- a) 3-methyl-6-oxo heptanal+++
- b) 3-methyl-1,6-heptanedione
- c) 2-keto-5-methyl-heptanal
- d) 5-methyl-2,7-heptanedione

78. What is responsible for the high boiling points of aldehydes and ketones?

- a) Hydrogen bonding existing between the molecules
- b) Intermolecular forces of attraction between molecules
- c) Polar interaction between molecules+++
- d) Dispersion forces between molecules

79. The Markovnikov rule states that a hydrogen atom bonds to the carbon atom that already is bonded to——— hydrogen atoms.

- a) an equal number of
- b) the greatest number of+++
- c) the least number of
- d) no

80. Carbocations undergo rearrangement reactions

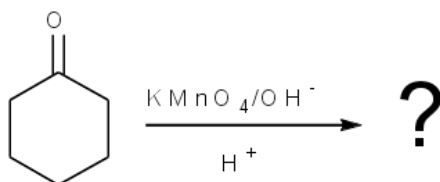
- a) to give more stable carbocations
- b) by hydride shifts
- c) by methyl shifts
- d) for all of the above reasons+++

81. Alkenes undergo an acid-catalyzed addition of water to give
- a) alkynes
 - b) alkanes
 - c) ethers
 - d) alcohols+++
82. Borane reacts with alkenes
- a) to give the anti-Markovnikov addition product
 - b) by a concerted mechanism
 - c) to give syn addition
 - d) all of the above+++
83. Diols are synthesized by reacting an alkene with
- a) water
 - b) borane
 - c) ozone
 - d) osmium tetroxide+++
84. Ozone reacts with 2,3-dimethyl-2-butene to give
- a) one ketone+++
 - b) two different ketones
 - c) one aldehyde
 - d) one ketone and one aldehyde
85. The rapid conversion between an enol and its keto form is called
- a) resonance
 - b) cationic rearrangement
 - c) tautomerization+++
 - d) hydrogenation
86. The solubility of some aldehydes and ketones in water is due to
- a) Hydrogen bonding between the molecules of water and those of aldehydes or ketones
 - b) Polar interaction and hydrogen bonding between the molecules of water and those of aldehydes or ketones+++
 - c) The intermolecular forces between the molecules of water and those of aldehydes or ketones
 - d) Dispersion and Van der Waal forces between the molecules of water and those of aldehydes or ketones

87. Aldehydes are said to be more reactive than ketones because
- The carbonyl carbon in the former is more polar than that of the latter
 - The carbonyl carbon in the latter is more sterically hindered from nucleophilic attack than in the former+++
 - The carbonyl carbon in the former is a terminal one hence better contact with nucleophiles unlike in the latter where the carbonyl carbon is not easily available for attack
 - The tetrahedral structure of the former possesses higher potential energy than the latter
88. Which of these has the most stable central carbon?
- RCOR' +++
 - RCOH
 - RCH_3
 - CH_4
89. The intermediate formed during the reaction of ozone with unsaturated compounds is called
- Oxidizing agent
 - Ozonide+++
 - Trioxane
 - Peroxide
90. Reaction of alkene with ozone in the presence of hydrogen peroxide will produce
- Two acids
 - Ketone and acid
 - Two ketones
 - All of the above+++
91. Ozonolysis is a very important reaction in chemistry because
- It is not toxic
 - It can be used to characterize complex molecules+++
 - It takes place at room temperature
 - It is a means of reducing complex molecules
92. Which of these is not a way of preparing aldehydes and ketones?
- Hydration of alkynes
 - Reduction of acid chlorides

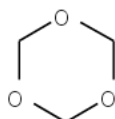
- c) Acylation reaction
- d) Amination reaction+++

93. What is the product of this reaction?



- a) Heptan-1,6-dioic acid+++
- b) Cycloheptan-1,6-dioic acid
- c) Ethanoic acid and pentane
- d) Cycloheptanoic acid

94. How many isomers will this compound have?



- a) 1
- b) 2
- c) 3+++
- d) 4

95. Which of these has the highest boiling point?

- a) Butane
- b) Butanal
- c) Butanol+++
- d) Butanone

96. What is the product you would get from the hydration of $\text{CH}_3\text{C}\equiv\text{CCH}_2\text{CH}_3$ using mercury salt?

- a) $\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_3$
- b) $\text{CH}_3\text{CH}_2\text{COCH}_2\text{CH}_3$
- c) $\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{COCH}_2\text{CH}_3$ +++
- d) $\text{CH}_3\text{CH}_2\text{C}(\text{OH})\text{CH}_2\text{CH}_3$

97. Aromatic ketones are prepared by

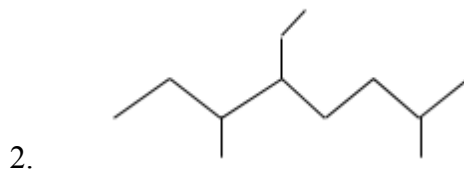
- a) Wittig reactions
- b) Friedel-Craft acylation+++

- c) Clemmensen reaction
 - d) Mercuration reaction
98. Cyanohydrins can be hydrolysed in the presence of acid to produce
- a) Compounds with both hydroxyl and carboxylic groups+++
 - b) Beta-Amino alcohols
 - c) Aldehydes
 - d) Ketones
99. What is the product of the reaction of water with ketones?
- a) Acetal
 - b) Hydrate+++
 - c) Hemiacetal
 - d) Geraniol
100. What is the final product of the reaction of alcohol with aldehyde called?
- a) Hemiacetal
 - b) Acetal+++
 - c) Hydrate
 - d) Ketal

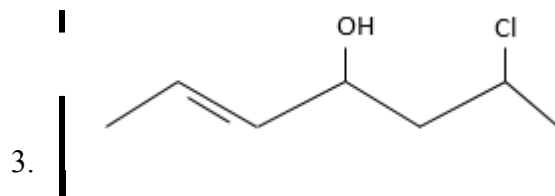
SECTION 2



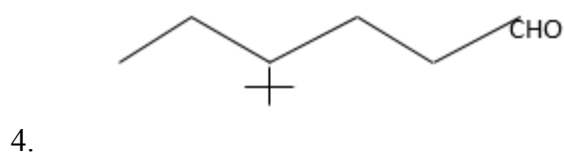
- A. Hex-1-en-6-yne
- B. Hex-1-en-1-yne
- C. Hept-1-en-6-yne+++
- D. Hept-6-en-1-yne



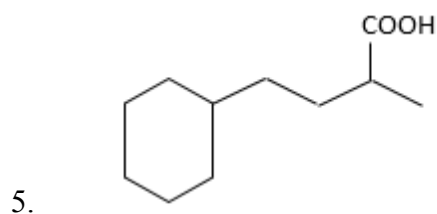
- A. 4-Ethyl-3, 7- dimethyl octane
- B. 5-Ethyl-2, 6-dimethyl octane+++
- C. 4-Ethyl-3, 6-dimethyl heptane
- D. 5-Ethyl-3, 7-dimethyl heptane



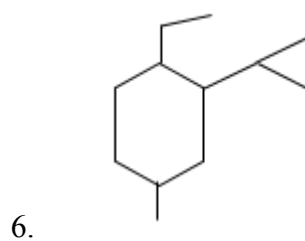
- A. 6-chloro hept-2-en-4-ol
- B. 2-chloro hept-5-en-4-ol+++
- C. 2-chloro hept-6-en-4-ol
- D. 2-chloro-4-hydroxy hept-5-ene



- A. 4-tertButyl hexanal+++
- B. 3-tertbutyl hexanal
- C. 4-Ethyl-5, 5-dimethyl hexanal
- D. 3-tertButyl hexanal

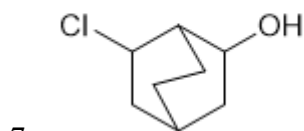


- A. 5-cyclotexyl carboxylic acid
- B. 3-cyclohexyl carboxylic acid
- C. 2-Ethyl-3-cyclohexyl butanoic acid
- D. 2-Methyl-4-cyclohexyl butanoic acid+++

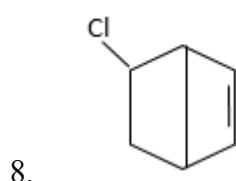


- A. 1-Ethyl-5- methyl-2-isopropyl cyclohexane
- B. 4- Ethyl-3-isopropyl-1-methyl cyclohexane
- C. 2- Ethyl-1-isopropyl-5-methyl cyclohexane

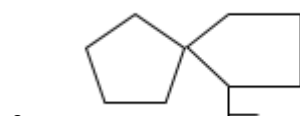
D. 1-Ethyl-2-isopropyl-4-methyl cyclohexane+++



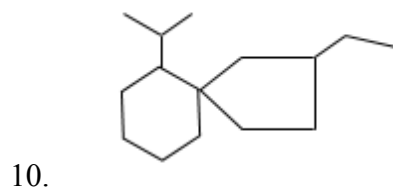
- A. 3-chloro -2-hydroxy bicyclo [2,2,2]-octane
- B. 6-chloro bicyclo [2,2,2] octan-2-ol+++
- C. 2-chloro bicyclo [2,2,2] octan-2-ol
- D. 5-chloro bicyclo [2,2,2] octan-3-ol



- A. 5-chloro bicyclo [2,2,0] hex-2-ene+++
- B. 2-chloro bicyclo [2,2,0] hex-5-ene
- C. 2-chloro bicyclo [2,2,0] hex-2-ene
- D. 5-chloro bicyclo [2,2,0] hex-1-ene



- A. 4-Ethyl spiro[4,5] nonane
- B. 1-Ethyl spiro[4,4]nonane+++
- C. 4-Ethyl sprio[4,4]octane
- D. 1-Ethyl spiro[5,5]nonane



- A. 2-isopropyl-8- Ethyl spiro [5,4] decane
- B. 2-Ethyl-6-isopropyl spiro [4,5] decane+++
- C. 3-Ethyl-7-isopropyl spiro [4,5] decane
- D. 3-Ethyl-10-isopropyl spiro [4,5] decane

11. hexene and 2-hexene are:

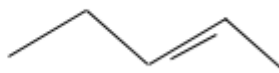
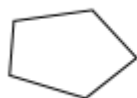
- A. Functional group isomers
- B. Chain isomers
- C. Alkene isomers
- D. Position isomers+++

12. These two compounds are:



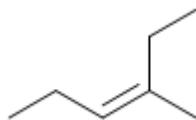
- A. Position isomers
- B. Functional group isomers+++
- C. Ether-alcohol isomers
- D. Chain isomers

13. These are:



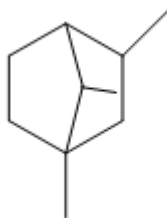
- A. Cycle-alkene isomers
- B. Functional group isomers
- C. Chain-cycle isomers+++
- D. Position isomers

14. The name of this compound is



- A. trans-2- methyl hex-3-ene
- B. Trans-3- methyl hex-3-ene
- C. Cis-3- methyl hex-3-ene+++
- D. Trans -2- methyl hex-2-ene

15. How many chiral centres are there in the structure?



- A. 1
- B. 2
- C. 3
- D. 4+++

16. An optical isomer has all except

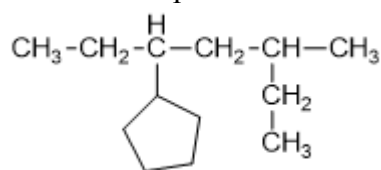
- A. has a chiral centre
- B. rotates plane polarised light

- C. has a mirror image
- D. is super imposable on its mirror image+++

17. A compound has 8 carbon atoms only six of the carbons are methyl radicals. The name of the compound is:

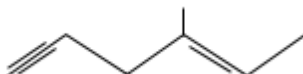
- A. 1,2-hexamethyl ethane
- B. 2,4-tetramethyl butane
- C. 2,2,3,3-tetramethyl butane+++
- D. 1,2,3,4-tetramethyl butane

18. Name the compound



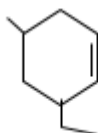
- A. 2-Ethyl-4- cyclopentyl hexane
- B. 3-Methyl-4- cyclopentyl heptane
- C. 3-Cyclopentyl-5-methylheptane+++
- D. 3- Methyl-6- cyclopentyl heptanes

19. Name the compound



- A. 4-Methyl hex-2-en-5-yne
- B. 4-Methylhex -1-yne-4-ene
- C. 4-methyl hex-4-en-1-yne+++
- D. 4-methyl hex-2-en-1-yne

20. This compound is



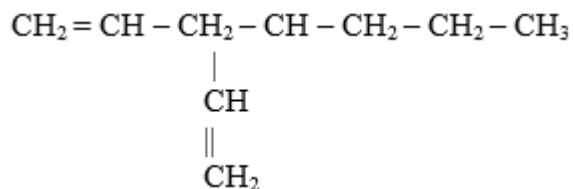
- A. 2-Ethyl-4-methyl cyclohexene
- B. 3-Ethyl-5-methyl cyclohexene+++
- C. 5-Ethyl-3-methyl cylohexene
- D. 1-Ethyl-5-methyl cyclohex-2-ene

21. Chain isomers are: compounds that have the same molecular

formula but differ in ---

- A. Optical arrangement
- B. Functional groups
- C. Spatial arrangement of atoms
- D. Bonding arrangement of radicals+++

22. Give the name of:



4-vinyl heptene

4-allylhexene

3-propylhexa-1, 5-diene+++

4-propyl hexa-1, 5-diene

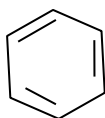
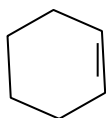
23. How many structural isomers can you write for C_5H_{12}

- A. 4
- B. 3+++
- A. 2
- B. 5

24. How many saturated cycles can you write from C_5H_{10}

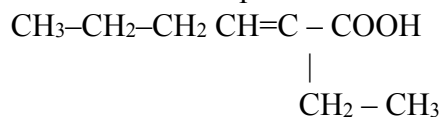
- A. 5
- B. 4
- C. 2
- D. 3+++

25. Which experiment will distinguish between



- A. Ignition
- B. Sodium
- C. Bromination+++
- D. Solubility test

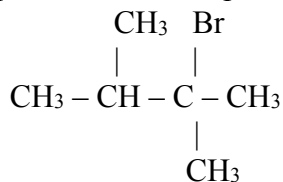
26. The suffix of the compound:



- A. Hexenoic acid
- B. Hex-2-enoic acid+++
- C. Hex-4-enoic acid

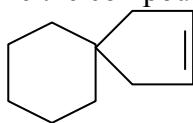
D. Hex-3-enoic acid

27. The prefix of this compound:



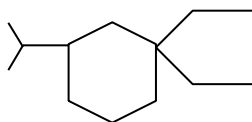
- A. 3-Bromo-1, 3-dimethyl
- B. 2,3-dimethyl-3-bromo
- C. 2-Bromo-2, 3-dimethyl+++
- D. 2-Bromo-2-Isoprophyl

28. Name the compound



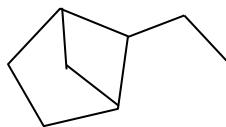
- A. Spiro {4,5} dec-2-ene+++
- B. Spiro {4,6} dec -2- ene
- C. Spiro {5,6} dec-2-ene
- D. Spiro {4,5} dec -1- ene

29. Pick the correct name for:



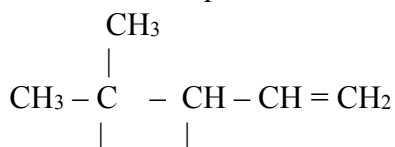
- A. Spiro {4,5} noriane
- B. 2-isopropyl spiro {4,5} nonane
- C. 3-isopropyl spiro {4,5} nonane
- D. 7-isopropyl spiro {4,5} nonane+++

30. This compound is:



- A. 1-ethyl bicyclo [2,1,1]- hexane
- B. 2-Ethyl bicyclo [211]- hexane+++
- C. 5-Ethyl bicyclo [2,1,1]- hexane
- D. 3-Ethyl bicycle [2,1,1] – hexane

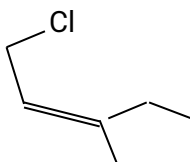
31. Name the compound





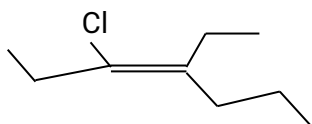
- A. 2,2 – Dimethyl-3-Vinyl pentene
- B. 2,2-Dimethyl-3-Vinyl pent-5-ene
- C. 3-tertbutylpenta-1, 4-diene+++
- D. 4,4-Dimethyl-3-Vinylpentene

32. The name of this compound is:



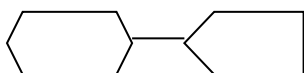
- A. 5-chloro-3-methyl pentene
- B. 5-chloro-3-methyl pent-3-ene
- C. 1-chloro-2-methyl pent-2-ene
- D. 1-chloro-3-methyl pent-2-ene+++

33. This compound is



- A. 3-chloro-4-ethyl hept-3-ene+++
- B. 3-chloro-3-ethyl hept-3-ene
- C. 4-chloro-3-propyl hex-3-ene
- D. 5-chloro-4-ethylhept-4-ene

34. What is the IUPAC name of the compound?



- A. Cyclohexane cyclopentane
- B. Cyclohexylcy clopentane
- C. Cyclopentyl cyclohexane+++
- D. Cyclopentane cyclohexane

35. The IUPAC name of this compound is



- A. Ethyl butyl ether
- B. Butyl ethyl ether
- C. Butoxy ethane
- D. Ethoxy butane+++

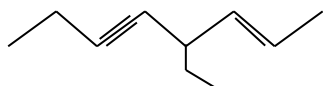
36. Which of these compounds has a chiral carbon?

- A. Chloropentane
- B. 3-chloropentane
- C. 2-chloropentane+++
- D. Chloro-2, 2-dimethylpropane

37. One of these compounds will not be optically active

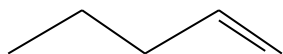
- A. Chlorobromomethane
- B. Chloro-2-methyl propane
- C. 2-chloro-2-methyl propane
- D. 2-chloropropane+++

38. What is the IUPAC name of this compound?

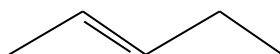


- A. 4-Ethyl oct-6-en-3-yne
- B. 4-Ethyl oct-2-en-5-yne+++
- C. 5-Ethyl oct-2-en-5-yne
- D. 4-Ethyl oct-2-en-3-yne

39.



a.

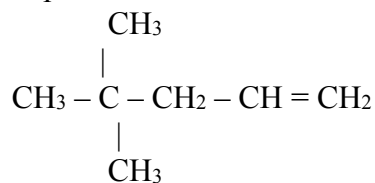


b.

a and b are:

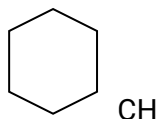
- A. Functional group isomers
- B. Tautomers
- C. Alkene isomers
- D. Position isomers+++

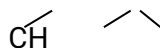
40. This compound is:



- A. 2,2-dimethyl pentene
- B. 4,4-dimethyl pentene+++
- C. 2,2-dimethyl pent-4-ene
- D. 4,4-dimethyl pent-4ene

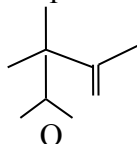
41. The name of this compound is:





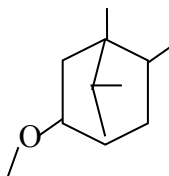
- A. 1,3-dimethyl hexane
- B. 1,3-dimethyl cyclohex-2-ene
- C. 1,3-dimethyl cyclohexene+++
- D. 1,3-dimethyl hex-1-ene

42. Name the compound below:



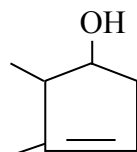
- A. 2-Methyl-2-isopropyl 1 butanone
- B. 2-Methyl-2-isopropyl butan-2-one
- C. 3,3,4-trimethyl pentan-3-one
- D. 3,3,4-trimethyl pentan-2-one+++

43. Name the Compound:



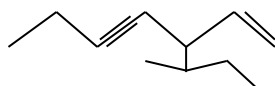
- A. 1,2,7-trimethyl-3-methoxy bicycle [2,2,10] heptane
- B. 1,2,7-trimethyl bicycle [2,2,1] heptanes
- C. 1,2,7-trimethyl-5-methoxy bicyclo [2,2,1] heptanes+++
- D. 1,2,7-trimethyl-4-methoxy bicycle [2,2,1] heptanes

44. What is the name of the compound?



- A. 1,3-dimethyl pent-3-enol
- B. 1,3-dimethylpent-2-enol
- C. 2,3-dimethyl pent-3-enol+++
- D. 1,3-dimethyl penten-3-ol

45. Give the name of the compound.

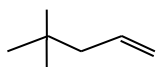


- A. 3 – secbutyl hept -1- en-4-yne+++
- B. 5-isopropyl hept-1-en-3-yne
- C. 3- isopropyl hept -6-en-3-yne
- D. 4-vingl-3-methyloet -3-yne

46. $\text{CH}_2=\text{CHCl}$ is

- A. Allylchloride
- B. Ethyl chloride
- C. Vinyl chloride+++
- D. Chloro ethylidene

47. Name the compound

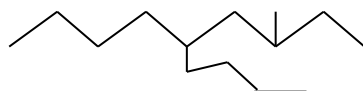


- A. 2,2-dimethyl pentene
- B. 4,4-dimethyl pentene+++
- C. 2,2-dimethyl pent-4-ene
- D. 4,4-dimethyl pent-2-ene

48. The different parts of a systematic name include all of these except

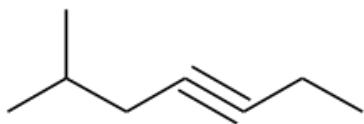
- A. suffix
- B. Principal chain
- C. Straight chain+++
- D. prefix

49. The prefix for this compound



- A. 3-Butyl-5-methyl.
- B. 5-butyl-7-methyl
- C. 3-methyl-4-butyl
- D. 5-butyl-3-methyl+++

50. The name of this compound is:

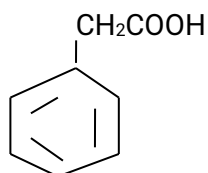


- A. 2-methylheptyne
- B. 2-methylhept-3-yne
- C. 2-methylhept-4-yne
- D. 6-methylhept-3-yne+++

51. The melting point of members of carboxylic acid homologous series
- A increases up the services
 - B decreases up the series
 - C the even carbon acids have higher melting points than their immediate higher and lower members+++
 - D show no definite trend in the series
52. The melting points of structural isomers of carboxylic acid series aretheir normal chain counterpart
- (A) Lower than+++
 - (B) higher than
 - (C) the same as
 - (D) not defined
53. Carboxylic acids have higher boiling points due to
- (A) the alkyl chain
 - (B) hydrogen bonding+++
 - (C) carboxyl group
 - (D) carboxylic acid bonding
54. Hell Volard Zelinsky reaction is
- (A) regioselective+++
 - (B) regiospecific
 - (C) stereoselective
 - (D) stereospecific
55. Hell-Volard Zelinsky reaction substitutes only
- (A) α -carbon atom
 - (B) β -carbon atom
 - (C) α -hydrogen+++

- (D) β -hydrogen
56. In Hell-Volard-Zelirsky a catalytic amount of.....is needed
- (A) sodium
 - (B) potassium
 - (C) red phosphorus+++
 - (D) bromonium ion
57. Which of these will reduce carboxylic acid to alcohol?
- (A) LiAlH_4 +++
 - (B) NaBH_4
 - (C) H_2/Pd
 - (D) H_2/Ni
58. ClCH_2COOH is more acidic than CH_3COOH by reason of
- (A) resonance effect
 - (B) conjugative effect
 - (C) cummulative effect
 - (D) inductive effect+++
59. When the hydrogen of the $-\text{COOH}$ is replaced by an alkyl group (R) it becomes (A) an amide
- (B) an ester+++
 - (C) a ketone
 - (D) an alkane
60. The end result of the reduction of carboxylic acid by borane (B_2H_4) in
- (A) an aldehyde
 - (B) a ketone
 - (C) an alcohol+++
 - (D) an alkane
61. Benzene can be differentiated from an alkene by the reaction
- (A) it burns with sooty lumminious flame
 - (B) it boils at 78°C
 - (C) it does not decolourise bromine water+++
 - (D) it melts at 5.5°C
62. Which of these groups will be good meta directing group
- (A) $-\text{NH}_2$

- (B) $-\text{CH}_3$
 (C) $-\text{NO}_2$ +++
 (D) $-\text{Cl}$
63. Benzene is insoluble in
 (A) chloroform
 (B) ethanol
 (C) water+++
 (D) acetone
64. Benzene can be produced from
 (A) ethylene
 (B) acetylene+++
 (C) methylene
 (D) propylene
65. Benzene is resistant to oxidation and reduction because double bonds are:
 (A) three
 (B) cumulated
 (C) additive
 (D) conjugated+++
66. The name of this compound



- (A) acetic benzene acid
 (B) benzyl acid
 (C) benzoic acid
 (D) 2-phenyl acetic acid+++
67. Benzene ring undergoes
 (A) substitution reactions +++
 (B) oxidation reaction
 (C) addition reaction
 (D) reduction reaction
68. Benzene is most available from this
 (A) petroleum+++

- (B) natural products
 - (C) wood
 - (D) saw dust
69. Benzene is oxidized readily with
- (A) ozone+++
 - (B) oxygen
 - (C) chlorine
 - (D) potassium permanganate
70. Acetophenone is easily prepared by
- (A) Friedel-Crafts benzoylation
 - (B) Friedel-Cryts alkylation
 - (C) Friedel-Crafts acylation+++
 - (D) phenylation of acetone

SECTION 3

1. The periodic table is a chart that arranges element according to
 - A. Atomic mass m
 - B. Atomic number Z +++
 - C. Molecular mass
 - D. None of the above
2. The principal use of diamond is in
 - A. Lighting
 - B. Jewelry +++
 - C. Lead pencil
 - D. Saw
3. The ability of metals to conduct electricity is due to
 - A. Surface tension
 - B. Smooth surface
 - C. Presence of free mobile electrons+++

- D. High tension
4. The reason why graphite is used as electrodes is due to;
- A. Absence of free mobile electrons
 - B. Presence of free mobile π electrons+++
 - C. Rigid structure
 - D. Flexible structure
5. Which of these elements is not alkaline - earth metal?
- A. Ba
 - B. Sr
 - C. Mg
 - D. Pd+++
6. Which of these elements is not an alkali metal?
- A. Li
 - B. Na
 - C. Ba+++
 - D. Cs
7. What is an alloy?
- A. A miscible mixture of two or three metals at their molten states+++
 - B. Two nonmetals
 - C. Nonmetals
 - D. Gases.
8. The Morseley's equation is;
- A. $E = mc^2$
 - B. $\sqrt{v} = a(Z - b) \quad +++$
 - C. $PV = mc^2$
 - D. $PV/T = \text{constant}$

9. Metals on ionization become charged with smaller

A. Sizes +++

B. Sites

C. Sources

D. Momentum

10. Diamond has which hybridization?

A. Sp

B. Sp^2

C. Sp^3 +++

D. Spd

11. Is graphite convertible to diamond?

A. Yes +++

B. No

C. Uncertain

D. None of the above

12. Electrolysis of water produces;

A. Hydrogen and oxygen +++

B. Nitrogen and coke

C. Colanut

D. Bitter cola

13. What is a crystal?

A. A solid with lattice points occupied by ions, atoms or molecule
+++

B. Glass

C. Cold room

D. Liquid

14. Example of amorphous solid is;

- A. Gold
- B. Glass +++
- C. Silver
- D. Mites

15. What is a metal?

- A. A malleable, ductile, lustrous solid that conducts electricity and heat. +++
- B. Insulator
- C. Perforator
- D. Plate

16. The characteristics of alkali metal compounds is that they are;

- A. White and form colourless solution in water +++
- B. Insoluble in water
- C. Black in colour
- D. None of the above

17. Which of these materials is not a covalent crystal?

- A. Copper+++
- B. Graphite
- C. Diamond
- D. Quartz

18. Period 1 contains;

- A. H and He+++
- B. Uranium and other actinides
- C. Lanthanides
- D. Halogens

19. Group 1 elements are;

- A. Copper, silver and gold

- B. Iron, manganese and cobalt
- C. Lithium, sodium, potassium, rubidium, caesium and francium
+++
- D. Hydrogen and helium

20. An isotope has;

- A. Element with two or more different atomic masses +++
- B. Element with different atomic number
- C. Element without mass
- D. Element with high electron affinity

21. Group 18 elements like Ar, He, Ne and Kr are known as;

- A. Noble gases+++
- B. Noble metals
- C. Precious stones
- D. Alkali

22. Sucrose, ice cube, sulphur, phosphorus and iodine are known as

- A. Metallic crystals
- B. Molecular crystal+++
- C. Jasmine
- D. Ionic crystals

23. Group 17 elements like I, Br, F, Cl are known as

- A. Hydrogen
- B. Halogens+++
- C. Petrogens
- D. Androgens

24. The lattice points of ionic crystals are occupied by;

- A. Molecules
- B. Tubers

C. Charged ions+++

D. Chronicles

25. Which of this is the property of diamond?

A. Hardness+++

B. Softness

C. Brittleness

D. Ductility

26. Metallic crystals contain;

A. Molecules

B. Charged ions of the metal in a sea or cloud of electrons +++

C. Butter

D. Yam

27. Electrovalent compounds are held together by;

A. Molecular forces

B. Electrovalent forces+++

C. Covalent forces

D. Military forces

28. The major use of graphite is;

A. Electrodes+++

B. Portal

C. Primer

D. Emerald

29. Diamond is used primarily as;

A. Jewelry+++

B. Pavement

C. Food

D. Farm

30. Which of these elements will liberate hydrogen from cold water?
- A. Phosphorus
 - B. Sulphur
 - C. Potassium+++
 - D. Purine
31. Electrolysis is
- A. Use of electrical current to break electrolytic compounds in molten or aqueous states to their constituents elements+++
 - B. Use of hammer to break substances
 - C. Use of adhesive to bind surfaces
 - D. Use of water to spray.
32. Which of these is not an alkali element?
- A. lithium
 - B. Gold+++
 - C. Sodium
 - D. Potassium
33. Which of these elements will displace hydrogen from cold water?
- A. Rubidium+++
 - B. Copper
 - C. Gold
 - D. Tungsten
34. The group in a periodic table shares;
- A. Similar electronic configuration physical and chemical properties+++
 - B. Different properties.
 - C. Different mobility
 - D. Different tools.
35. What force holds covalent compounds together?

- A. Ionic
- B. Van der Waal+++
- C. Cubic
- D. Rectangular

36. Which of these are allotropes of carbon?

- A. Diamond and graphite+++
- B. Procter and Gamble
- C. Silver
- D. Solar

37. Gold is a

- A. Bullet
- B. Malt
- C. Metal+++
- D. Glass

38. Which of these is the characteristic of Transition elements?

- A. Partly filled d-orbitals+++
- B. Fully filled life cycle
- C. Non metals
- D. Polarizable electrodes

39. Which of these is not the characteristic of lanthanides and actinide?

- A. Non metals+++
- B. Radioactive
- C. Partially filled f-orbitals
- D. Synthetic elements

40. Which of these elements is not a transition metal?

- A. Iron
- B. Manganese

C. Zinc

D. Fluorine+++

41. Which of these elements is not a halogen?

A. Fluorine

B. Chlorine

C. Iodine

D. Boron+++

42. Which of these substances is not a good conductor of electricity?

A. Copper

B. Silver

C. Iron

D. Diamond+++

43. Why is graphite a good conductor of electricity?

A. It contains mobile π electrons+++

B. It is rigid

C. It flows

D. It is a gas.

44. When an element with high electron affinity gains an electron to become an anion; its size;

A. Decreases+++

B. Does not change

C. Increases

D. Not certain.

45. The ions in ionic crystal lattice have;

A. Same size

B. Different sizes+++

C. Unknown sizes

D. Large sizes.

46. Which of these substances is amorphous?

A. Glass+++

B. Sodium chloride

C. Calcium nitrate

D. Diamond

47. The molecular crystals have this property;

A. Melt at below 100°C +++

B. Very hard solid

C. Flows

D. Fluid

48. Can diamond be converted to graphite?

A. Yes

B. No+++

C. Not certain

D. Possible

49. Halogens are highly reactive because they;

A. Have high electro affinity+++

B. Low ionization energy

C. Low boiling point

D. Inert

50. The reactivity of alkali elements is due to;

A. Low ionization energy+++

B. High electron affinity

C. Inertness

D. Not sure

51. Mendeleev's Periodic Table was based on:

- A. Atomic mass+++
- B. Ionization energy.
- C. Atomic radius
- D. Ionic radius.

52. NaCl is an ionic compound because

- A. Ions are held in lattice+++
- B. It is covalent molecule
- C. The melting point is below freezing point.
- D. It is formed by sharing of electrons

53. Morseley's Periodic Table was rather based on:

- A. Covalent radius
- B. Atomic number++++
- C. Electron sharing
- D. Electrode potentials

54. Metal structure has a sea of delocalized electrons functioning for:

- A. Conduction of electricity+++
- B. Covalent bonding
- C. Molecule
- D. Electron sharing.

55. Why are members of the same group having similar physical and chemical properties?

- A. They have similar electronic configuration+++
- B. They conduct heat
- C. They are able to conduct electricity
- D. None at all

56. Alkali metals are highly electropositive because they

- A. Hold tightly their outermost electrons.

- B. Easily release their outermost electrons for ionization+++
- C. Have moisture.
- D. They are covalent.

57. Ice, Sulphur and iodine are examples of:

- A. Molecular crystals+++
- B. Solid with high melting points.
- C. No lattice
- D. None of the above.

58. Graphite, diamond and quartz are examples of:

- A. Brittle materials
- B. Strongly covalent crystals+++
- C. No rigid structures
- D. Crude solids

59. Diamond has:

- A. sp^3 hybridisation+++
- B. Sp hybridization
- C. Spd hybridization
- D. Spdf hybridization.

60. Graphite is mostly used as:

- A. Glass cutter
- B. Electrode+++
- C. Jewelry
- D. Brittle material.

61. Diamond is an important:

- A. Jewelry+++
- B. Lead pencil
- C. Lubricant

D. Braille.

62. Which of these elements is not an alkali metal?

- A. Gold+++
- B. Sodium
- C. Caesium
- D. Rubidium

63. Which of these elements is not an alkaline earth metals?

- A. Sulphur+++
- B. Calcium
- C. Barium
- D. Magnesium.

64. Covalent compounds are held together by:

- A. van der waal forces+++
- B. Collagen
- C. Gum
- D. Ionic forces.

65. Chlorine, fluorine, bromine, iodine and astatine belong to the group of:

- A. Chalcogen
- B. Halogen+++
- C. Transition metals
- D. Inner transition metals.

66. Lanthanides and Actinides have:

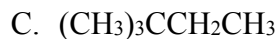
- A. Partially filled outermost f-orbitals+++
- B. No orbital.
- C. S-orbital only
- D. None of the above.

67. Noble gases have the following properties:

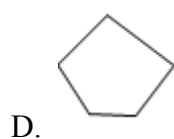
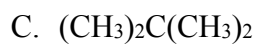
- A. Vacancies in their in their s-orbitals
 - B. No vacancies in their outermost s- or p-orbitals+++
 - C. Vacancies in p-orbitals
 - D. Vacancies in d-orbitals.
68. Elements in Period 7 in the Periodic table are unstable with short half-lives because they are:
- A. Synthetic and elements++++
 - B. Naturally occurring
 - C. Stable
 - D. None of the above
69. Helium, xenon, neon, krypton and argon in Group 18 are called:
- A. Noble gases++++
 - B. Chalcogens
 - C. Halogens
 - D. Actinides
70. Is graphite convertible to diamond?
- A. Not sure
 - B. No
 - C. Yes+++
 - D. Uncertain

SECTION 4

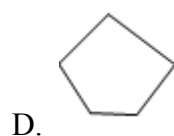
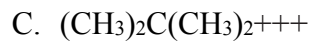
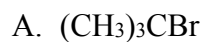
24. Which of the following alkanes has the lowest boiling point?
- A. $(\text{CH}_3)_2\text{CHCH}(\text{CH}_3)_2$
 - B. $(\text{CH}_3\text{CH}_2)_2\text{CHCH}_3$
 - C. $(\text{CH}_3)_3\text{CCH}_2\text{CH}_3$ +++
 - D. $\text{CH}_3(\text{CH}_2)_4\text{CH}_3$
25. Which of the following alkanes has the highest boiling point?



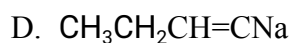
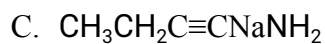
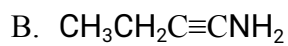
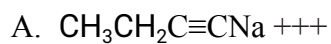
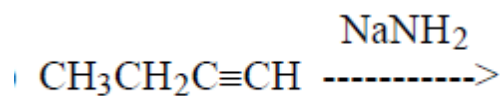
26. Which of the following compounds has the highest boiling point?



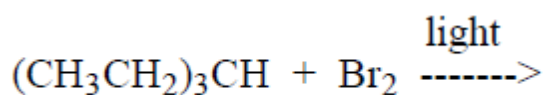
27. Which of the following compounds has the lowest boiling point?



28. Predict the product or products of this reaction.

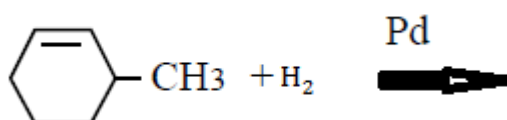


29. Predict the product or products of this reaction.



- A. $(\text{CH}_3\text{CH}_2)_3 \text{Br} + \text{HBr}$
- B. $(\text{CH}_3\text{CH}_2)_3 \text{CBr} + \text{HBr} \text{ +++}$
- C. $(\text{CH}_3\text{CH}_2)_3 \text{HBr} + \text{HBr}$
- D. $(\text{CH}_3\text{CH}_2)_2 \text{Br} + \text{HBr}$

30. Predict the product or products of this reaction



- A. +++
- B.
- C.
- D.

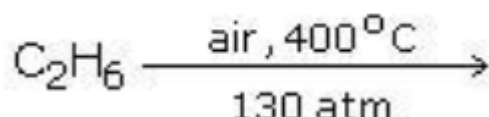
31. What are hydrocarbons?

- A. Compounds containing only the elements carbon called hydrocarbons
- B. Compounds containing only the elements carbon and oxygen are called hydrocarbons
- C. Compounds containing the elements carbon, hydrogen and oxygen are called hydrocarbons
- D. Compounds containing only the elements carbon and hydrogen are called hydrocarbons +++

32. Choose one which does not belong to this set: C_2H_2 , C_3H_4 , C_3H_6

- A. C_3H_6 +++
- B. C_2H_2

- C. C_3H_6
D. C_3H_4
33. An organic compound A with the formula C_2H_6O gives the compound B with the same number of carbon atoms when oxidized, which further undergoes oxidation to acetic acid. Identify the compound A.
- A. Acid
B. Ethanol+++
C. Aldehyde
D. Ketone
34. What are the products of reaction between ethanol and concentrated sulphuric acid at $170^\circ C$?
- A. Ethene
B. Propene and water
C. Ethene and water+++
D. Butene
35. In a molecule of a hydrocarbon, the number of 'C' atoms is '5'. What will be its formula if it is an alkyne?
- A. C_5H_{10}
B. C_5H_{12}
C. C_5H_8 +++
D. C_5H_{14}
36. What would be the product of reaction this reaction below



- A. Ethane
B. C. Benzene
C. C. Acetic acid+++
D. D. Ethanol
37. Which of the following will not give addition reactions; Propane, propyne, propene, benzene.
- A. Propane+++
B. Propyne
C. Propene
D. Benzene
38. Which one of the following compounds is not expected to be completely soluble in water at room temperature.



- B. CH_3COOH
- C. $\text{CH}_3(\text{CH}_2)_{16}\text{COOH}$
- D. $\text{CH}_3\text{+++}$

39. The functional group of a molecule does not affect a compound's

- A. solubility
- B. melting point
- C. reaction with other molecules
- D. carbon chain length+++

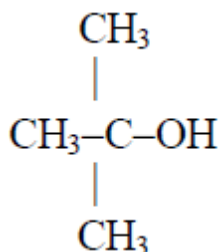
40. The combustion of a hydrocarbon will produce mandatory:

- A. SO_2
- B. $\text{CO}_2\text{+++}$
- C. CO
- D. SO_3

41. Which of the following substances presents only primary carbon atoms in its structure?

- A. methane
- B. acetylene
- C. ethane+++
- D. propane

42. The formula of the compound below represents



- A. primary alcohol
- B. secondary alcohol
- C. tertiary alcohol+++
- D. quaternary alcohol

43. What can result by the reduction of an aldehyde?

- A. a primary alcohol+++
- B. a secondary alcohol
- C. a carboxylic acid
- D. a compound with a mixed function alcohol – aldehyde

44. What can result by the reduction of a ketone?

- A. a primary alcohol
- B. a secondary alcohol+++
- C. a carboxylic acid
- D. a compound with a mixed function alcohol – ketone

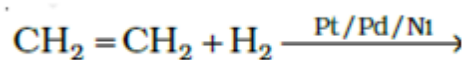
45. The alcoholic fermentation is a reaction of:

- A. dehydration
- B. hydrolysis
- C. esterification

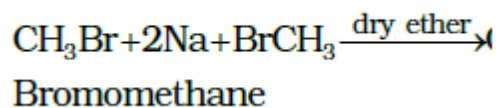
- D. catalyzed enzymatically+++
46. H₂ does not react with any of the isomers of the following molecular formulas:
- C₂H₄
 - C₂H₂
 - C₆H₆
 - C₄H₁₀+++
47. In which of the following reactions a C-C bond is broken:
- $\text{CH}_3\text{-CN} + 2\text{H}_2 \longrightarrow \text{CH}_3\text{-CH}_2\text{-NH}_2$
 - $\text{C}_6\text{H}_5\text{-CH}_3 + 3\text{H}_2 \longrightarrow \text{C}_6\text{H}_{11}\text{-CH}_3$
 - $\text{C}_6\text{H}_5\text{-NO}_2 + 3\text{H}_2 \longrightarrow \text{C}_6\text{H}_5\text{-NH}_2 + 2\text{H}_2\text{O}$
 - $\text{C}_6\text{H}_5\text{-CH}_3 + \text{H}_2 \longrightarrow \text{C}_6\text{H}_6 + \text{CH}_4$ +++
48. Organic compounds can contain other elements such as oxygen or nitrogen. True or False +++
- A. True
49. Organic compounds are always derived from a living source. True or False +++
- A. False
50. An unlimited number of carbon atoms can bond together to form chains. True or False +++
- A. True
51. Most organic compounds contain ionic bonds. True or False +++
- A. False
52. There are more inorganic compounds than organic compounds. True or False +++
- A. False+++
53. Most inorganic compounds have a higher melting point than organic compounds. True or false +++
- A. True
54. Inorganic compounds are never found in living systems. True or false +++
- A. False
55. Isomers always have the same empirical (molecular) formula. True or false +++
- A. True
56. Isomers have identical physical and chemical properties. True or false +++
- A. False
57. Carbon-carbon bonds are usually strong and covalent. True or false +++
- A. True
58. A characteristic of the alkanes is that they burn in air. True or false +++
- A. True
59. What is the reason why there are so many organic compounds?
- Organic compounds are produced by living things
 - There are cyclic organic compounds, and none of

- the inorganic compounds are cyclic
- C. Carbon atoms can form stable covalent bonds with other carbon atoms+++
- D. Organic compounds can be larger than inorganic compound
60. Which statement referring to functional groups is correct?
Functional groups are
- A. where nothing happens in a chemical reaction
- B. the parts of molecules used to classify organic molecules
- C. where most chemical reaction occur
- D. both B and C+++
61. Identify the correct statement: The alkanes represent one family of organic compounds composed of carbon compounds
- A. in which the halogens are a major component
- B. in which there is at least one double bond between carbons
- C. that are composed only of carbon and hydrogen+++
- D. in which hydrogens can be bonded to other hydrogens
62. Which of the following is an alcohol?
- A. $\text{CH}_3\text{CH}_2\text{COOH}$
- B. $\text{CH}_3\text{CH}_2\text{OH}$ +++
- C. CH_3COCH_3
- D. $\text{CH}_3\text{CH}_2\text{Br}$
63. What is meant by the term "saturated" when describing hydrocarbons?
- A. The hydrocarbons are saturated with water and are wet.
- B. The hydrocarbons are saturated with carbons in its parent chain.
- C. The hydrocarbons are saturated with hydrogens and have only single bonds between carbons+++
- D. The compound is pure
64. Which of the following formulas represents a saturated hydrocarbon?
- A. C_2H_6 +++
- B. C_2H_4
- C. C_3H_6
- D. C_5H_{10}
65. What is a characteristic shared by the alkanes?
- A. solids
- B. polar
- C. flammable+++
- D. colored
66. Which of the following compounds exhibits structural isomerism?

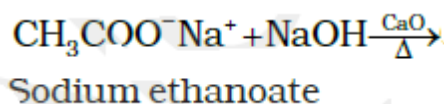
- A. propane
 - B. methane
 - C. butane+++
 - D. ethane
67. An alkane with 10 carbon atoms is a _____ at room temperature.
- A. solid
 - B. liquid+++
 - C. gas
 - D. not possible to tell
68. Alkanes react readily with
- A. A sodium hydroxide
 - B. hydrochloric acid
 - C. sulfuric acid
 - D. none of the above+++
69. What would be the product (s) of this reaction?



- A. Ethane +++
 - B. Ethanol
 - C. Ethyne
 - D. None of the above
70. What would be the product (s) of this reaction?



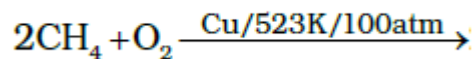
- A. $\text{CH}_3 - \text{CH}_3$
 - B. $\text{CH}_3 - \text{CH}_2 - \text{Br}$
 - C. $\text{CH}_3 - \text{CH}_2 - \text{NaBr}$
 - D. $\text{CH}_3 - \text{CH}_3 + 2\text{NaBr}$ +++
71. What would be the product (s) of this reaction?



- A. CH_4

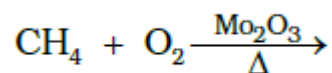
- B. CH_3COOH
- C. $\text{CH}_4 + \text{NaOH}$
- D. $\text{CH}_4 + \text{Na}_2\text{CO}_3^{+++}$

72. What would be the product (s) of this reaction?



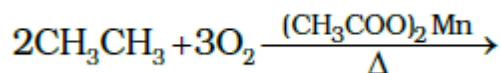
- A. $\text{H}_2\text{O} + \text{CO}_2$
- B. CH_3OH
- C. $2\text{CH}_3\text{OH}^{+++}$
- D. CH_3COOH

73. What would be the product (s) of this reaction?



- A. HCHO
- B. $\text{H}_2\text{O} + \text{CO}_2$
- C. CH_3OH
- D. $\text{HCHO} + \text{H}_2\text{O}^{+++}$

74. What would be the product (s) of this reaction?



- A. $\text{CH}_3\text{COOH} + \text{H}_2\text{O}$
- B. $2\text{CH}_3\text{COOH} + 2\text{H}_2\text{O}^{+++}$
- C. $\text{CH}_3\text{CH}_2\text{OH}$
- D. $\text{CH}_3\text{CH}_2\text{COOH}$

75. CH_3Cl can be used in the manufacture of

- A. silicones
- B. methyl cellulose
- C. Synthetic rubber
- D. All⁺⁺⁺

76. Which of the following can be used as local anesthetic?

- A. Chloroethane⁺⁺⁺
- B. Trichlorofluoromethane
- C. Chloroform
- D. Bromotrifluoromethane

77. The first three members are gases, the next fourteen are

liquids and the higher ones are solids. True or false +++

A. True

78. Straight chain alkenes have higher boiling point than isomeric branched chain compounds. True or false +++

A. True

79. This reaction is possible in the presence of the catalyst



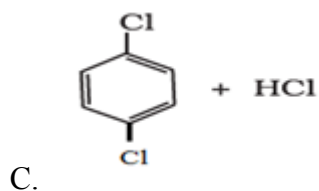
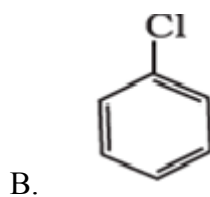
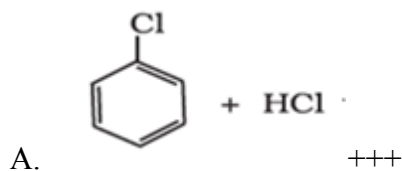
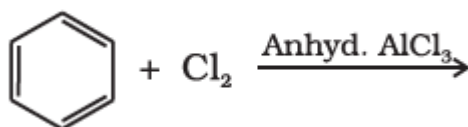
A. Pt/ Ni

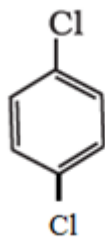
B. Ni/Pd

C. Pt/Pd/ Ni +++

D. Cu/ Pt/ Pd

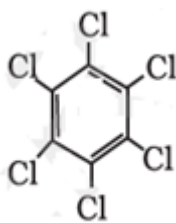
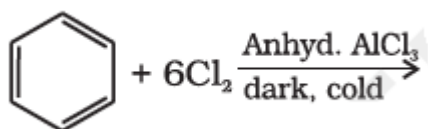
80. What would be the product (s) of this reaction?





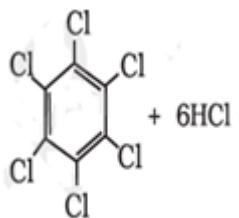
D.

81. What would be the product (s) of this reaction?

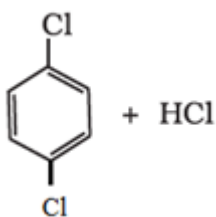


A.

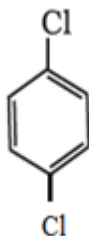
+++



B.



C.



D.

82. Boiling point aryl group is in the following order: $\text{Ar} - \text{I} > \text{Ar} - \text{Br} > \text{Ar} - \text{Cl} > \text{Ar} - \text{F}$. True or False +++

A. True

SECTION 5

1. What are the products of this reaction; $2\text{Na} + 2\text{H}_2\text{O}$?
 - A. $2\text{NaOH} + \text{H}_2$ +++
 - B. $\text{N}_2 + \text{HNO}_3$
 - C. $\text{NO}_2 + \text{O}_2 + \text{BaO}$
 - D. None of the above.
2. The periodic table is a chart that arranges element according to:
 - A. Atomic mass m.
 - B. Atomic number Z. +++
 - C. Molecular mass.
 - D. None of the above.
3. The principal use of diamond is in;
 - A. Lighting.
 - B. Jewelry. +++
 - C. Lead pencil
 - D. Saw.
4. The principal use of graphite is in;
 - A. Lighting.
 - B. Jewelry.
 - C. Lead pencil +++
 - D. Saw.
5. The ability of metals to conduct electricity is due to;
 - A. Surface tension.
 - B. Presence of mobile ions
 - C. Presence of mobile electrons. +++
 - D. High tension.
6. The reason why graphite is used as electrodes is due to;

- A. Absence of free mobile electrons.
 - B. Presence of free mobile π electrons. +++
 - C. Rigid structure.
 - D. Flexible structure.
7. Which of these elements is not alkaline - earth metal?
- A. Ba
 - B. Sr
 - C. Mg
 - D. Pd +++
8. Which of these elements is not an alkali metal?
- A. Li
 - B. Na
 - C. Ba +++
 - D. Cs
9. Which of these are not the properties of hydrogen?
- A. Conduction of electricity. +++
 - B. Reducing agent.
 - C. Hardening of oil.
 - D. Liberation of copper from copper salt.
10. Hydrogen can be produced in any of these groups except
- A. Group i
 - B. Group iv
 - C. Group vii
 - D. Group iii +++
11. What is a halogen?
- A. Member of group 19 or 0 in the periodic table.
 - B. Most electronegative elements. +++
 - C. Member of group 1.

D. Member of group 2.

12. Which of the following elements is not a halogen?

A. H. +++

B. Cl

C. I

D. Br

13. Balance this equation; $\text{CH}_3\text{COOH} + \text{NaOH}$

A. $\text{CH}_3\text{COONa} + \text{H}_2\text{O}$. +++

B. $\text{NaCl} + \text{H}_2\text{O}$

C. K_2O

D. $\text{MgCl}_2 + \text{H}_2\text{O}$

14. What is the principal use of lithium salts?

A. Treatment of certain mental disorder. +++

B. Battery acid.

C. Commercial train.

D. Fertilizers.

15. Potassium sulphate and phosphate are used as;

A. Battery electrolyte

B. Electrodes

C. Fertilizers +++

D. Fermentation.

16. The major use of bromine is in;

A. Production of insecticides. +++

B. Isolation of metals.

C. Finding of warms.

D. Grilling of meat.

17. Calcium carbide is used in;

A. Calcinations.

- B. Production of acetylene used for welding purposes. +++
- C. Rat poisoning.
- D. Chrome plating

18. Complete the following reactions; $2\text{Na} + \text{Cl}_2 \rightarrow$

- A. 2NaCl +++
- B. BaSO_4
- C. MgO
- D. PbSO_4

19. The products of pyrolysis of 2NaNO_3 are;

- A. $2\text{NaNO}_2 + \text{O}_2$ +++
- B. NaCl
- C. KCl
- D. PbO .

20. Sodium and potassium are best prepared by;

- A. Hydrolysis of their salts.
- B. Electrolysis of their eutectic mixtures. +++
- C. Reaction of their oxides with carbon.
- D. Reduction of their oxides with hydrogen.

21. Complete this reaction $\text{Zn} + 2\text{HCl} \rightarrow$

- A. $\text{ZnO} + \text{O}_2$
- B. $\text{MgO} + \text{H}_2$
- C. $\text{ZnCl}_2 + \text{H}_2$ +++
- D. $\text{ZnO} + \text{HCl}$

22. Complete this reaction; $2\text{Na} + 2\text{NH}_3 \rightarrow$

- A. NaH_2
- B. NaCl
- C. NaHg
- D. $\text{NaNH}_2 + \text{H}_2$ +++

23. Radium is not stable because it is;
- A. Radioactive with short half-life. +++
 - B. Very stable.
 - C. Dormant.
 - D. Ionic
24. The product of the pyrolysis of $\text{Ca}(\text{NO}_3)_2$ are;
- A. Ammonia
 - B. Oxide
 - C. $\text{CaO} + \text{NO}_2 + \text{O}_2$. +++
 - D. Barite
25. The reaction between $\text{Ca}(\text{HCO}_3)_2$ and 2HCl produces;
- A. $\text{CaCl}_2 + \text{H}_2\text{O} + \text{CO}_2$ +++
 - B. CaSO_4
 - C. Ca_3PO_4
 - D. MnO
26. Metals on ionization become charged with smaller
- A. Sizes +++
 - B. Sites
 - C. Sources
 - D. Momentum
27. Ionization energy decreases along the period
- A. True
 - B. False +++
 - C. Neither true nor false
 - D. None of the above
28. Ionization energy increases along the period
- A. True +++
 - B. False

C. Neither true nor false

D. None of the above

29. Atomic radius decreases along the period

A. True +++

B. False

C. Neither true nor false

D. None of the above

30. Atomic radius increases along the period

A. True

B. False +++

C. Neither true nor false

D. None of the above

31. Ionization energy decreases down the group

A. True +++

B. False

C. Neither true nor false

D. None of the above

32. Ionization energy increases down the group

A. True

B. False +++

C. Neither true nor false

D. None of the above

33. Atomic radius decreases down the group

A. True

B. False +++

C. Neither true nor false

D. None of the above

34. Atomic radius increases down the group

- A. True +++
- B. False
- C. Neither true nor false
- D. None of the above

35. Atomic radius remains unchanged down the group

- A. True
- B. False +++
- C. Neither true nor false
- D. None of the above

37. Sodium chloride is less soluble in water than in solution of sodium sulphate

- A. True
- B. False +++
- C. Neither true nor false
- D. None of the above

38. Potassium chloride is more soluble in water than in solution of sodium chloride

- A. True +++
- B. False
- C. Neither true nor false
- D. None of the above

39. Balance the equation $2\text{LiH} + \text{H}_2\text{O} \rightarrow$

- A. Li_3N
- B. LiF
- C. LiCl
- D. $\text{LiOH} + \text{H}_2$ +++

40. Balance the equation, $\text{Na}_2\text{O} + \text{H}_2\text{SO}_4 \rightarrow$

- A. $\text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$ +++
- B. NaOH
- C. NaCl

D. KCl

41. Balance the equation $\text{CaC}_2 + 2\text{H}_2\text{O} \rightarrow$

A. $\text{C}_2\text{H}_2 + \text{Ca}(\text{OH})_2$ +++

B. CaO

C. CaOCl

D. $\text{Ca}_3(\text{PO}_4)_2$

42. Mention the use of CaCO_3 ;

A. Manufacture of cement. +++

B. Blending of salt.

C. Cure of hepatitis.

D. Production of acetylene.

43. The most abundant element in the earth crust is

A. Oxygen +++

B. Argon

C. Neon

D. Krypton

44. Lithium hydride is an example of;

A. Interstitial hydride

B. Ionic hydride +++

C. Covalent hydride

D. None of the above.

45. Lithium hydride is an example of;

A. Interstitial hydride

B. Ionic hydride

C. Covalent hydride +++

D. None of the above.

46. Which of these is not a hydride;

A. Methane.

- B. LiH
- C. MgH_2
- D. CaCl_2 +++

47. Diamond has which hybridization?

- A. Sp
- B. Sp^2
- C. Sp^3 +++
- D. Spd

48. Graphite has which hybridization?

- A. Sp
- B. Sp^2 +++
- C. Sp^3
- D. Spd

49. What is the major source of francium?

- A. Nuclear reactor +++
- B. Nitrogen
- C. Gallium
- D. Cocaine

50. Barite is the source of;

- A. Barium +++
- B. Strontium
- C. Silver
- D. Gold

51. Is graphite convertible to diamond?

- A. Yes +++
- B. No
- C. Uncertain
- D. None of the above.

52. Electrolysis of water produces;

- A. Hydrogen and oxygen +++
- B. Nitrogen and coke
- C. Colanut
- D. Bitter cola

53. Balance this equation; $\text{Sr}(\text{OH})_2 + 2\text{HNO}_3 \rightarrow$

- A. $\text{Sr}(\text{NO}_3)_2 + 2\text{H}_2\text{O}$ +++
- B. BaSO_4
- C. BaO .
- D. Phosphorus.

54. Balance the reaction's equation; $\text{CaC}_2 + \text{N}_2 \rightarrow$

- A. Chlorine
- B. Argon
- C. $\text{Ca}(\text{CN})_2$ +++
- D. Methane

55. What is a crystal?

- A. A solid with lattice points occupied by ions, atoms or molecule +++
- B. Glass
- C. Cold room
- D. Liquid

56. Example of amorphous solid is;

- A. Gold
- B. Glass +++
- C. Silver
- D. Mites

57. The product of this reaction $\text{Ca} + 2\text{C}$ is;

- A. CaC_2 +++
- B. Mallot

C. Margot

D. Mould

58. What is a metal?

A. A malleable, ductile, lustrous solid that conducts electricity and heat.

+++

B. Perforator

C. Plate

D. Insulator

59. The characteristics of alkali metal compounds is that they are;

A. White and form colourless solution in water. +++

B. Insoluble in water

C. Black in colour

D. None of the above

60. Hydrogen belongs to group vii because of the presence of one electron in its outermost shell

A. True

B. False +++

C. Neither true nor false

D. None of the above